Explainable Double DQN for Ultrafiltration Control in Hemodialysis

Selected Topics: Reinforcement Learning & Explainable AI

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Project idea

An RL agent learns how fast to remove water during dialysis; Integrated Gradients shows why it chose each action.

1

Simulate

Real patient profiles go into a small simulator. One session = 16 steps of 15 minutes.

2

Learn

A Double DQN agent tries water-removal rates and learns from the reward.

3

Explain

Integrated Gradients shows which inputs pushed each choice.

Problem statement: an RL problem

Medical context: dialysis removes extra water from the blood. Removing it too fast makes blood pressure fall.

16 decisions in a row

The agent picks a rate every 15 minutes, 16 times per session.

Delayed reward

A fast rate now can drop blood pressure some steps later.

A trade-off

Remove the water, but keep blood pressure safe.

Why not a fixed rule? It uses one rate all session and does not react when blood pressure falls.

Why look inside? A neural-network policy is a black box. We must see which inputs drive each choice.

Literature review

Paper	What it gives our RL work
Yang et al. 2022 [1]	RL (Duel-DDQN) for a dialysis decision: dry weight
van Hasselt et al. 2016 [2]	Double DQN: our learning algorithm
Sundararajan et al. 2017 [3]	Integrated Gradients: our explanation method
Li et al. 2025 [4]	Explainable model that warns of blood-pressure problems
Wang et al. 2025 [5]	Predicts low blood pressure, but no control
Kanbay et al. 2020 [6]	Rules for our simulator and the 13 mL/kg/h limit
Cheungpasitporn et al. 2026 [9]	Review: AI and RL in kidney care
Tang et al. 2026 [10]	Review of RL for treatment: clear S, A, R; need for explanations

The gap: papers [4, 5] only predict risk. No paper explains a learned controller during the session. We do that with DDQN + IG.

Proposed approach: environment and data

UCI CKD data [7, 8]

400 real patient profiles

→

Simulator

4 simple equations

→

RL agent

learns by trial and error

320 training patients

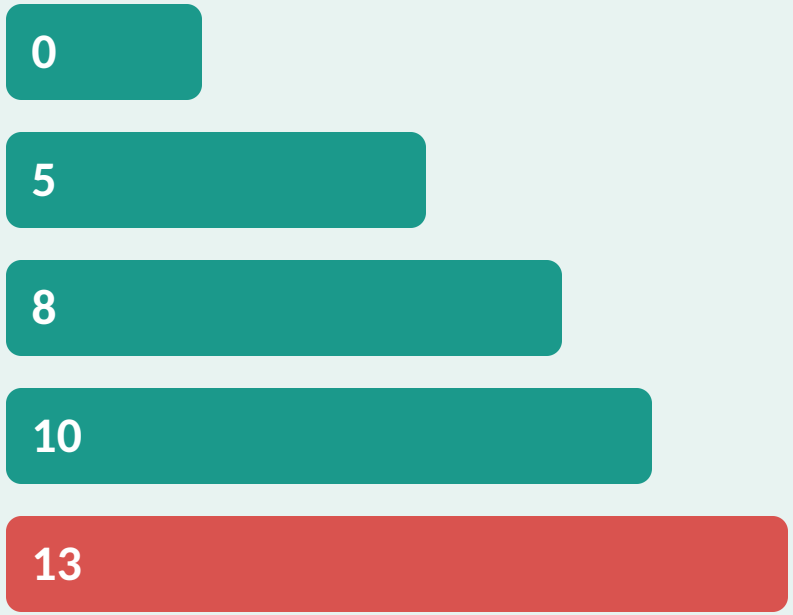
80 test patients (never seen in training)

- From the data: age, diastolic blood pressure, urine albumin. The ckd label is not used.
- Why this data: real, open (CC BY 4.0) and widely used; no public dialysis time series exists.

MDP: state S and action A

State feature	Plain meaning
overload	water still to remove (kg)
removed	water removed so far (kg)
sbp	systolic blood pressure (top number)
dbp	diastolic blood pressure (bottom number)
hr	heart rate
time	how far into the session (0 to 1)
age	patient age
albumin	blood protein that helps water refill

Action A: UF rate (mL/kg/h)



0 = stop. 13 = highest safe rate [6].

The agent sees the state, picks an action, and gets a reward. Then it sees the next state.

MDP: transition S → S′ and reward R

(1) $dv = \min(u \cdot W \cdot \Delta t \,/\, 1000, \text{overload});\ \text{overload}' = \text{overload} - dv$

Water removed this step = rate × weight × time.

(2) $SBP' = SBP - a\cdot(1 + age_f)\cdot(u - refill) + b\cdot(SBPo - SBP) + noise$

Removing water faster than the body refills lowers blood pressure.

(3) $DBP' = DBP + 0.5\cdot(SBP' - SBP) + noise$

The bottom pressure follows the top pressure.

(4) $HR' = HRo + c\cdot(SBPo - SBP') + noise$

The heart beats faster when pressure falls.

Reward $R = dv - 10\cdot[SBP' < 90] - 2\cdot\text{overload}$ (last step only)

+ for water removed · −10 for each low-pressure step · −2 per kg of water left at the end

From Q-learning to Double DQN

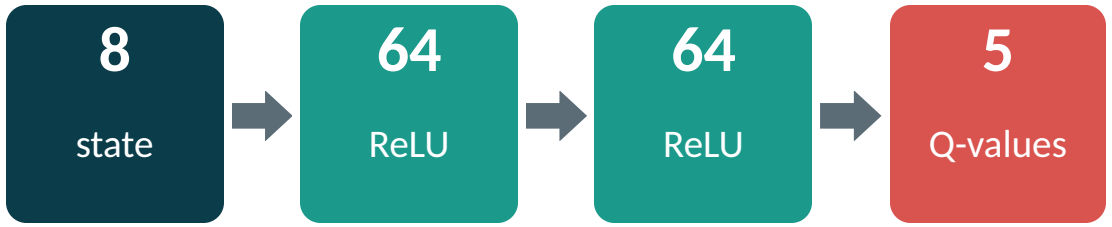
1	**Q(s, a)**	Expected total reward if we take action a in state s, then act well.
2	**Q-learning**	$Q(s, a) \leftarrow r + \gamma \cdot \max Q(s', a')$ (Bellman update)
3	**DQN**	A table cannot store continuous states, so a neural network learns Q.
4	**Problem**	The same network picks and scores the next action, so Q is too high (overestimation).
5	**Double DQN**	The online network picks the action; the target network scores it.

DQN target: $y = r + \gamma \cdot \max_a' Q_target(s', a')$

Double DQN: $y = r + \gamma \cdot Q_target(\ s', argmax_a' Q_online(s', a')\ )$

γ (gamma) = how much future reward counts. Picking and scoring with two networks stops the too-high values [2].

Our DDQN agent



In: the 8 state numbers. Out: one Q-value for each of the 5 rates. The agent takes the biggest.

Part	Our value	Why it is there
Replay buffer	50,000 steps, batch 64	Random old steps, so one batch is not one session
ε-greedy	1.0 → 0.05 over first 60% of 1500 episodes	Explore first, then use what it learned
Target network	copied every 200 learning steps	A stable goal to learn toward
Discount γ	0.99	Future reward counts almost as much as now
Optimizer	Adam, lr 1e-3, Huber loss, grad clip 10	Steady updates; big errors do not explode

Training loop

```
for seed in 0..4:
  for episode in 1..1500:
    s = reset()              # random training patient
    ε = decay(episode)       # 1.0 → 0.05
    repeat 16 steps:
      a = random if rand < ε else argmax Q(s)
      s', r = step(a)
      buffer.store(s, a, r, s')
      batch = buffer.sample(64)
      update Q_online on DDQN target
      every 200 updates: Q_target ← Q_online
      s = s'
```

5 seeds (0–4)

1500 episodes per seed

16 steps per episode

24,000 steps per seed

Proposed solution: baselines and why

Rule-based

No learning. One constant rate per patient to remove the water in 4 h, max 13 mL/kg/h.

What if we do not learn?

Tabular Q-learning

Classic RL with a table. State cut into bins: 5 water × 5 BP × 16 time steps. 3000 episodes.

Do we need a network?

Standard DQN

Same network and settings as ours, but without the Double fix.

What does the Double fix add?

Double DQN (ours)

Online network picks, target network scores.

Our method.

Proposed solution: the pipeline



If the check fails, main.py stops before IG.

- Why check first? Explaining a model that has not learned well tells us nothing.
- The approval is tied to the exact model file (SHA-256 hash = a fingerprint of the file).
- All seeds are fixed, so the results can be repeated.

```
$ python main.py
```

Results: test patients

Controller	Mean reward ↑	Sessions with low BP ↓	Low-BP steps per session ↓	Water left (kg) ↓	Rate jump ↓
DDQN (ours)	**1.022 ± 0.251**	**1.0% ± 1.5**	**0.010 ± 0.015**	**0.545 ± 0.090**	**1.62 ± 0.56**
Standard DQN	1.027 ± 0.106	1.8% ± 1.5	0.024 ± 0.022	0.496 ± 0.056	2.37 ± 0.47
Tabular Q-learning	−0.710 ± 0.595	2.8% ± 0.7	0.184 ± 0.066	0.542 ± 0.035	3.92 ± 0.15
Rule-based (13 cap)	−41.574	50%	4.40	0.110	0.00

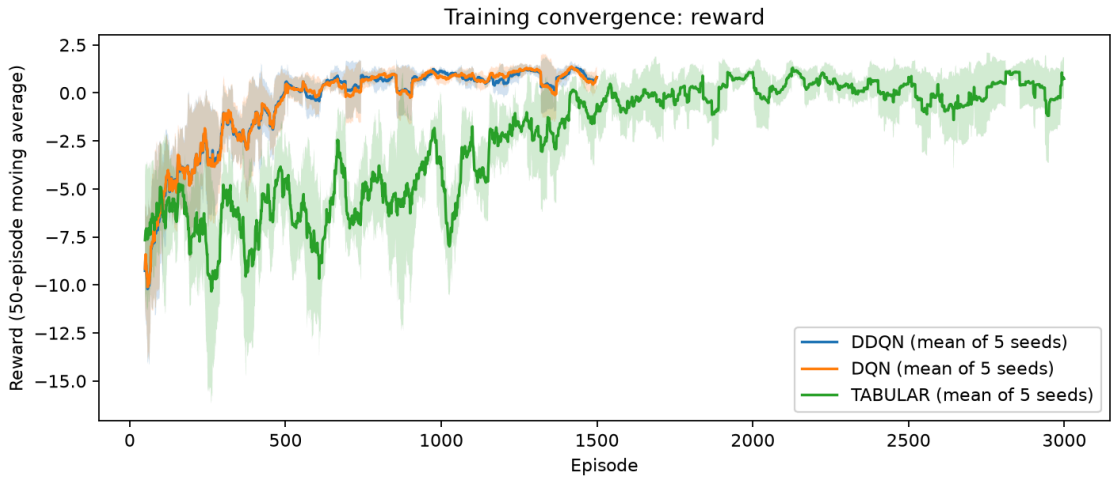
100 test sessions per seed, mean ± std over 5 seeds. Rate jump = average change of the UF rate between steps (mL/kg/h).

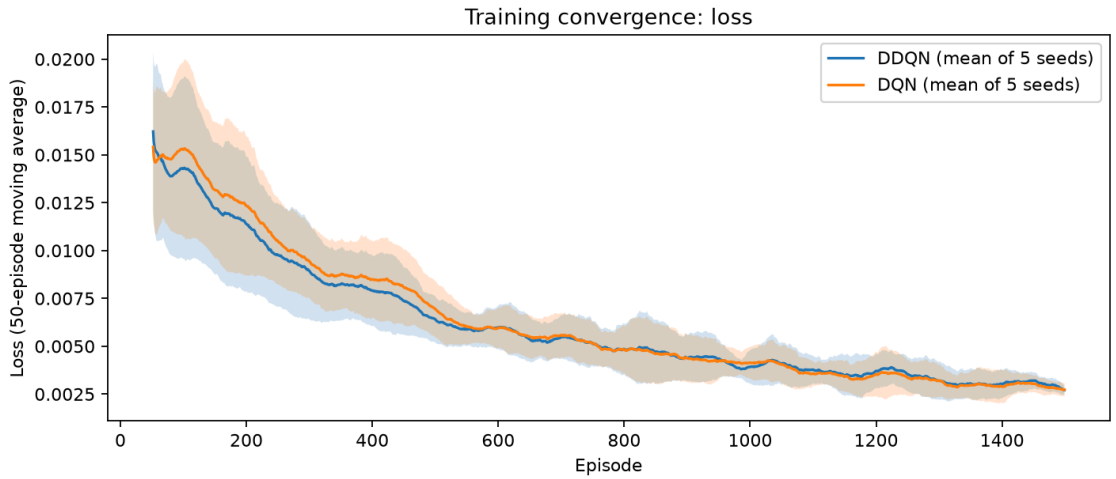
≈ DDQN and DQN: same reward

1.0% vs 1.8% low-BP sessions

50% rule: low BP in half the sessions

Learning curves and checks (before IG)





✓ Converged

Rewards stopped changing, on all 5 seeds

✓ Stable

5 seeds agree: spread 0.25 < 0.5

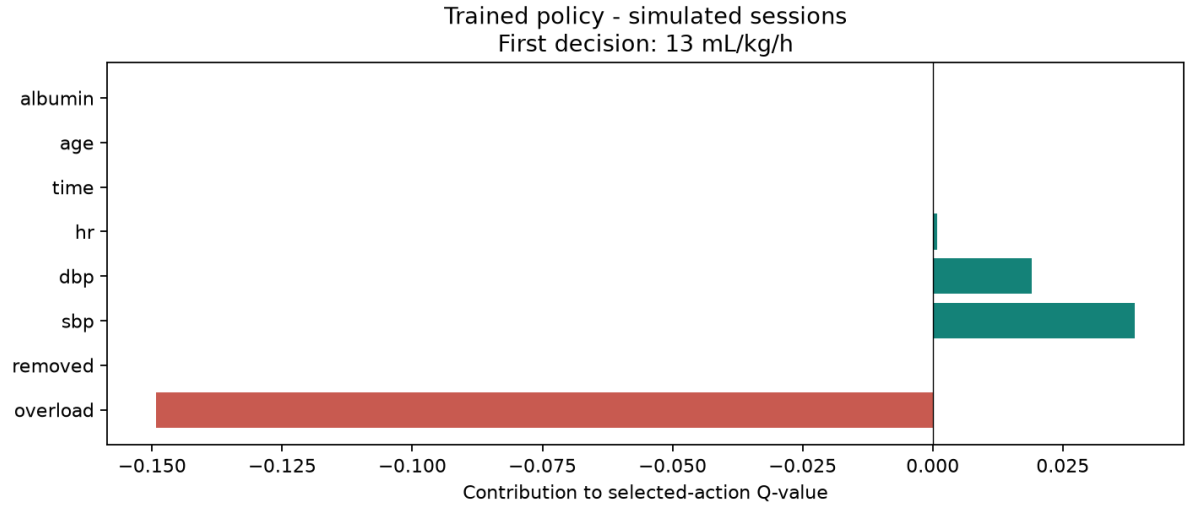
✓ Safe

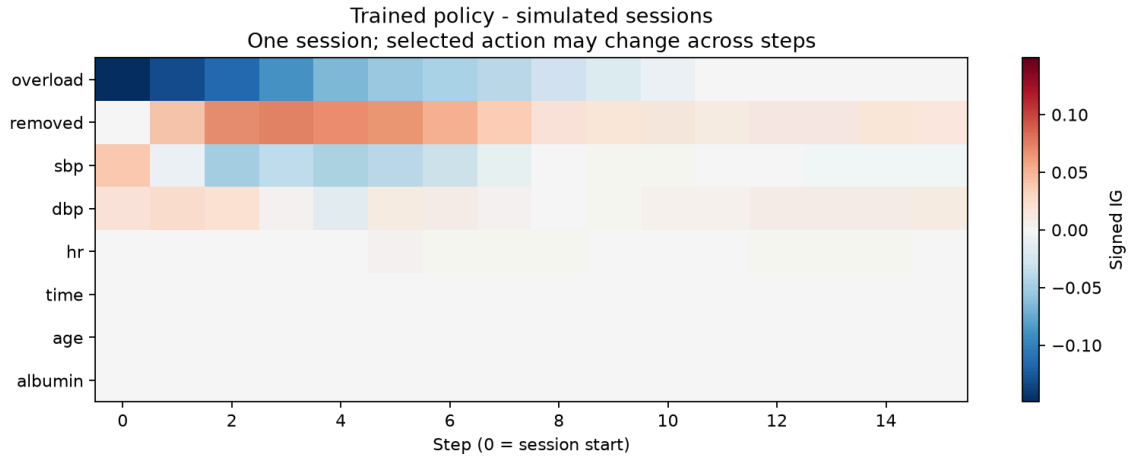
Few unsafe sessions: worst seed 4% ≤ 5%

APPROVED for IG

DDQN, 5 seeds, 100 test sessions per seed (results/validation_report.md)

Integrated Gradients, simply





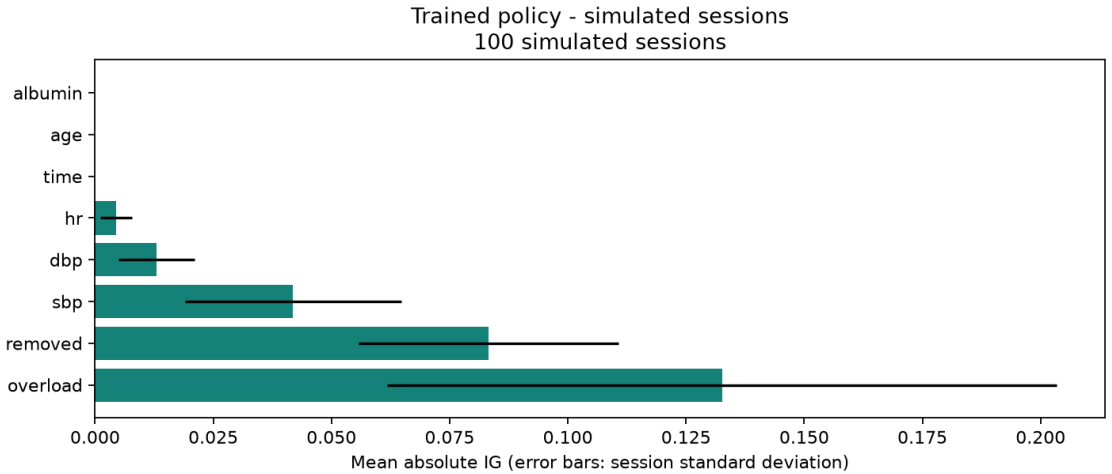
$IG_i = (x_i - b_i) \cdot \int_0^1 \partial Q_a / \partial x_i \, d\alpha$ [3]
= how much input i pushed the chosen action, compared with a reference patient b.

Top: one decision (rate 13)
Water left pulls it down; blood pressure pushes it up.

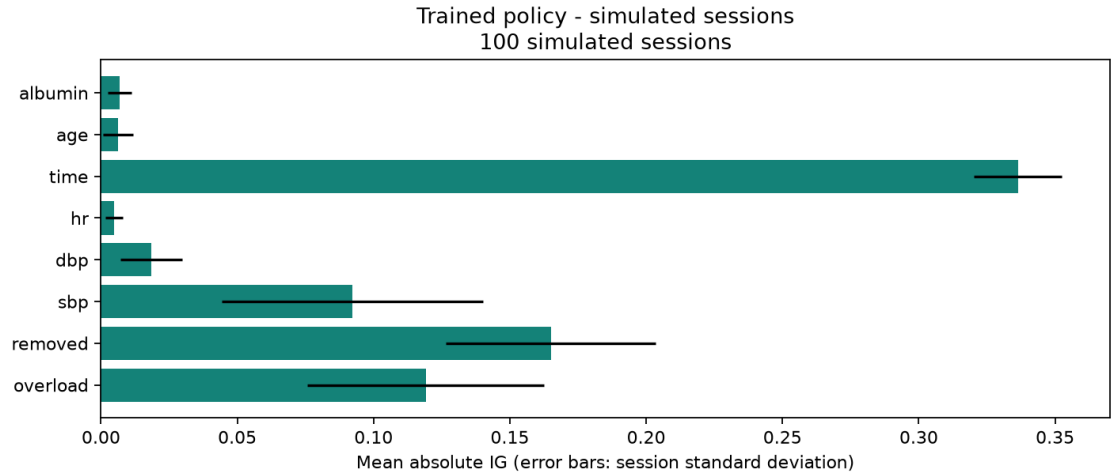
Bottom: one session
Water left matters most early, then fades.

The parts add up to the Q-value change (checked on every decision).

IG across 1600 decisions



Reference 1: default (no extra water)



Reference 2: average training patient

mean |IG|	overload	removed	sbp	dbp	hr	time	age	albumin
Default	0.133	0.083	0.042	0.013	0.004	0*	0*	0*
Average	0.119	0.165	0.092	0.018	0.005	0.336	0.006	0.007

** zero by design: this reference keeps time, age and albumin equal to the patient's own values*

The agent uses water left, water removed and blood pressure most; age and albumin barely matter.

Discussion

What we found

- DDQN ≈ DQN on reward (1.022 vs 1.027)
- DDQN is safer: 1.0% vs 1.8% low-BP sessions
- DDQN is smoother: rate jump 1.62 vs 2.37
- Fixed rule: most water removed, but unsafe in 50% of sessions

Limits

- Simple, hand-tuned simulator, not real sessions
- IG sessions use all 400 profiles, not only test patients
- IG is not cause and effect
- IG results depend on the reference patient

Bottom line: same reward as DQN, but safer and smoother, in simulation only.

Future work

1

Prioritized replay

Learn more often from steps with big errors.

2

Dueling network

Split Q into state value + action advantage.

3

More seeds + tuning

Search learning rate, γ, buffer size, ε schedule.

4

Offline RL on real data

Learn from logged sessions (Lin et al. 2019).

5

Contrastive IG

Explain why action A and not action B.

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